

31.7-2

Prove that if Alice's public exponent e is 3 and an adversary obtains Alice's secret exponent d , where $0 < d < \phi(n)$, then the adversary can factor Alice's modulus n in time polynomial in the number of bits in n . (Although you are not asked to prove it, you might be interested to know that this result remains true even if the condition $e = 3$ is removed. See Miller [327].)

★ 31.7-3

Prove that RSA is multiplicative in the sense that

$$P_A(M_1)P_A(M_2) = P_A(M_1M_2) \pmod{n}.$$

Use this fact to prove that if an adversary had a procedure that could efficiently decrypt 1% of messages from $\mathbb{Z}_n$ encrypted with P_A , then the adversary could employ a probabilistic algorithm to decrypt every message encrypted with P_A with high probability.

★ 31.8 Primality testing

This section shows how to find large primes. We begin with a discussion of the density of primes, proceed to examine a plausible, but incomplete, approach to primality testing, and then present an effective randomized primality test due to Miller and Rabin.

The density of prime numbers

Many applications, such as cryptography, call for finding large “random” primes. Fortunately, large primes are not too rare, so that it is feasible to test random integers of the appropriate size until you find one that is prime. The *prime distribution function* $\pi(n)$ specifies the number of primes that are less than or equal to n . For example, $\pi(10) = 4$, since there are 4 prime numbers less than or equal to 10, namely, 2, 3, 5, and 7. The prime number theorem gives a useful approximation to $\pi(n)$.

Theorem 31.37 (Prime number theorem)

$$\lim_{n \rightarrow \infty} \frac{\pi(n)}{n / \ln n} = 1 .$$

■

The approximation $n / \ln n$ gives reasonably accurate estimates of $\pi(n)$ even for small n . For example, it is off by less than 6% at $n = 10^9$, where $\pi(n) = 50,847,534$ and $n / \ln n \approx 48,254,942$. (To a number theorist, 10^9 is a small number.)

The process of randomly selecting an integer n and determining whether it is prime is really just a Bernoulli trial (see Section C.4). By the prime number theorem, the probability of a success—that is, the probability that n is prime—is approximately $1 / \ln n$. The geometric distribution says how many trials must occur to obtain a success, and by equation (C.36) on page 1197, the expected number of trials is approximately $\ln n$. Thus, in order to find a prime that has the same length as n by testing integers chosen randomly near n , the expected number examined would be approximately $\ln n$. For example, the expectation is that finding a 1024-bit prime would require testing approximately $\ln 2^{1024} \approx 710$ randomly chosen 1024-bit numbers for primality. (Of course, to cut this figure in half, choose only odd integers.)

The remainder of this section shows how to determine whether a large odd integer n is prime. For notational convenience, we assume that n has the prime factorization

$$n = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r} ,$$

where $r \geq 1$, $p_1, p_2, \dots, p_r$ are the prime factors of n , and $e_1, e_2, \dots, e_r$ are positive integers. The integer n is prime if and only if $r = 1$ and $e_1 = 1$.

One simple approach to the problem of testing for primality is *trial division*: try dividing n by each integer $2, 3, 5, 7, 9, \dots, \lfloor \sqrt{n} \rfloor$, skipping even integers greater than 2. We can conclude that n is prime if and only if none of the trial divisors divides n . Assuming that each trial division takes constant time, the worst-case running time is $\Theta(\sqrt{n})$, which is

exponential in the length of n . (Recall that if n is encoded in binary using β bits, then $\beta = \lceil \lg(n + 1) \rceil$, and so $\sqrt{n} = \Theta(2^{\beta/2})$.) Thus, trial division works well only if n is very small or happens to have a small prime factor. When it works, trial division has the advantage that it not only determines whether n is prime or composite, it also determines one of n 's prime factors if n is composite.

This section focuses on finding out whether a given number n is prime. If n is composite, we won't worry about finding its prime factorization. Computing the prime factorization of a number is computationally expensive. You might be surprised that it turns out to be much easier to ascertain whether a given number is prime than it is to determine the prime factorization of the number if it is not prime.

Pseudoprimality testing

We'll start with a method for primality testing that "almost works" and, in fact, is good enough for many practical applications. Later on, we'll refine this method to remove the small defect. Let $\mathbb{Z}_n^+$ denote the nonzero elements of $\mathbb{Z}_n$:

$$\mathbb{Z}_n^+ = \{1, 2, \dots, n-1\}.$$

If n is prime, then $\mathbb{Z}_n^+ = \mathbb{Z}_n^*$.

We say that n is a *base- a pseudoprime* if n is composite and

$$a^{n-1} \equiv 1 \pmod{n}. \tag{31.39}$$

Fermat's theorem (Theorem 31.31 on page 932) implies that if n is prime, then n satisfies equation (31.39) for every a in $\mathbb{Z}_n^+$. Thus, if there is any $a \in \mathbb{Z}_n^+$ such that n does *not* satisfy equation (31.39), then n is certainly composite. Surprisingly, the converse *almost* holds, so that this criterion forms an almost perfect test for primality. Instead of trying every value of $a \in \mathbb{Z}_n^+$, test to see whether n satisfies equation (31.39) for just $a = 2$. If not, then declare n to be composite by returning COMPOSITE. Otherwise, return PRIME, guessing that n is prime (when, in fact, all we know is that n is either prime or a base-2 pseudoprime).

The procedure PSEUDOPRIME on the next page pretends in this manner to check whether n is prime. It uses the procedure MODULAR-EXPONENTIATION from Section 31.6. It assumes that the input n is an odd integer greater than 2. This procedure can make errors, but only of one type. That is, if it says that n is composite, then it is always correct. If it says that n is prime, however, then it makes an error only if n is a base-2 pseudoprime.

How often does PSEUDOPRIME err? Surprisingly rarely. There are only 22 values of n less than 10,000 for which it errs, the first four of which are 341, 561, 645,

PSEUDOPRIME(n)

```
1 if MODULAR-EXPONENTIATION(2,  $n - 1$ ,  $n$ )  $\neq 1 \pmod{n}$ 
2   return COMPOSITE // definitely
3 else return PRIME  // we hope!
```

and 1105. We won't prove it, but the probability that this program makes an error on a randomly chosen β -bit number goes to 0 as β approaches ∞ . Using more precise estimates due to Pomerance [361] of the number of base-2 pseudoprimes of a given size, a randomly chosen 512-bit number that is called prime by PSEUDOPRIME has less than one chance in 10^{20} of being a base-2 pseudoprime, and a randomly chosen 1024-bit number that is called prime has less than one chance in 10^{41} of being a base-2 pseudoprime. Thus, if you are merely trying to find a large prime for some application, for all practical purposes you almost never go wrong by choosing large numbers at random until one of them causes PSEUDOPRIME to return PRIME. But when the numbers being tested for primality are not randomly chosen, you might need a better approach for testing primality. As we'll see, a little more cleverness, and some randomization, will yield a primality-testing method that works well on all inputs.

Since PSEUDOPRIME checks equation (31.39) for only $a = 2$, you might think that you could eliminate all the errors by simply checking equation (31.39) for a second base number, say $a = 3$. Better yet, you

could check equation (31.39) for even more values of a . Unfortunately, even checking for several values of a does not eliminate all errors, because there exist composite integers n , known as *Carmichael numbers*, that satisfy equation (31.39) for *all* $a \in \mathbb{Z}_n^*$. (The equation does fail when $\gcd(a, n) > 1$ —that is, when $a \notin \mathbb{Z}_n^*$ —but demonstrating that n is composite by finding such an a can be difficult if n has only large prime factors.) The first three Carmichael numbers are 561, 1105, and 1729. Carmichael numbers are extremely rare. For example, only 255 of them are less than 100,000,000. Exercise 31.8-2 helps explain why they are so rare.

Let's see how to improve the primality test so that Carmichael numbers won't fool it.

The Miller-Rabin randomized primality test

The Miller-Rabin primality test overcomes the problems of the simple procedure PSEUDOPRIME with two modifications:

- It tries several randomly chosen base values a instead of just one base value.
- While computing each modular exponentiation, it looks for a nontrivial square root of 1, modulo n , during the final set of squarings. If it finds one, it stops and returns COMPOSITE. Corollary 31.35 from Section 31.6 justifies detecting composites in this manner.

The pseudocode for the Miller-Rabin primality test appears in the procedures MILLER-RABIN and WITNESS. The input $n > 2$ to MILLER-RABIN is the odd number to be tested for primality, and s is the number of randomly chosen base values from $\mathbb{Z}_n^+$ to be tried. The code uses the random-number generator RANDOM described on page 129: RANDOM(2, $n - 2$) returns a randomly chosen integer a satisfying $2 \leq a \leq n - 2$. (This range of values avoids having $a \equiv 1 \pmod{n}$.) The call of the auxiliary procedure WITNESS(a, n) returns TRUE if and only if a is a “witness” to the compositeness of n —that is, if it is possible using a to prove (in a manner that we will see) that n is composite. The test WITNESS(a, n) is an extension of, but more

effective than, the test in equation (31.39) that formed the basis for PSEUDOPRIME, using $a = 2$.

Let's first understand how WITNESS works, and then we'll see how the Miller-Rabin primality test uses it. Let $n - 1 = 2^t u$ where $t \geq 1$ and u is odd. That is, the binary representation of $n - 1$ is the binary representation of the odd integer u followed by exactly t zeros. Therefore, $a^{n-1} = (a^u)^{2^t}$, so that one way to compute $a^{n-1} \bmod n$ is to first compute $a^u \bmod n$ and then square the result t times successively.

```

MILLER-RABIN( $n, s$ )           //  $n > 2$  is odd
1  for  $j = 1$  to  $s$ 
2     $a = \text{RANDOM}(2, n - 2)$ 
3    if WITNESS( $a, n$ )
4      return COMPOSITE // definitely
5  return PRIME           // almost surely

WITNESS( $a, n$ )
1  let  $t$  and  $u$  be such that  $t \geq 1$ ,  $u$  is odd, and  $n - 1 = 2^t u$ 
2   $x_0 = \text{MODULAR-EXPONENTIATION}(a, u, n)$ 
3  for  $i = 1$  to  $t$ 
4     $x_i = x_{i-1}^2 \bmod n$ 
5    if  $x_i == 1$  and  $x_{i-1} \neq 1$  and  $x_{i-1} \neq n - 1$ 
6      return TRUE           // found a nontrivial square root
                              // of 1
7  if  $x_t \neq 1$ 
8    return TRUE           // composite, as in
                              PSEUDOPRIME
9  return FALSE

```

This pseudocode for WITNESS computes $a^{n-1} \bmod n$ by first computing the value $x_0 = a^u \bmod n$ in line 2 and then repeatedly squaring the result t times in the **for** loop of lines 3–6. By induction on i ,

the sequence $x_0, x_1, \dots, x_t$ of values computed satisfies the equation $x_i = a^{2^i}$ for $i = 0, 1, \dots, t$, so that in particular $x_t = a^{n-1} \pmod{n}$. After line 4 performs a squaring step, however, the loop will terminate early if lines 5–6 detect that a nontrivial square root of 1 has just been discovered. (We'll explain these tests shortly.) If so, the procedure stops and returns TRUE. Lines 7–8 return TRUE if the value computed for $x_t = a^{n-1} \pmod{n}$ is not equal to 1, just as the PSEUDOPRIME procedure returns COMPOSITE in this case. Line 9 returns FALSE if lines 6 or 8 have not returned TRUE.

The following lemma proves the correctness of WITNESS.

Lemma 31.38

If WITNESS(a, n) returns TRUE, then a proof that n is composite can be constructed using a as a witness.

Proof If WITNESS returns TRUE from line 8, it's because line 7 determined that $x_t = a^{n-1} \pmod{n} \neq 1$. If n is prime, however, Fermat's theorem (Theorem 31.31) says that $a^{n-1} = 1 \pmod{n}$ for all $a \in \mathbb{Z}_n^*$. Since $\mathbb{Z}_n^+ = \mathbb{Z}_n^*$ if n is prime, Fermat's theorem also says that $a^{n-1} = 1 \pmod{n}$ for all $a \in \mathbb{Z}_n^+$. Therefore, n cannot be prime, and the equation $a^{n-1} \pmod{n} \neq 1$ proves this fact.

If WITNESS returns TRUE from line 6, then it has discovered that x_{i-1} is a nontrivial square root of 1, modulo n , since we have that $x_{i-1} \neq \pm 1 \pmod{n}$ yet $x_i = x_{i-1}^2 = 1$. Corollary 31.35 on page 934 states that only if n is composite can there exist a nontrivial square root of 1, modulo n , so that demonstrating that x_{i-1} is a nontrivial square root of 1, modulo n proves that n is composite. ■

Thus, if the call WITNESS(a, n) returns TRUE, then n is surely composite, and the witness a , along with the reason that the procedure returns TRUE (did it return from line 6 or from line 8?), provides a proof that n is composite.

Let's explore an alternative view of the behavior of WITNESS as a function of the sequence $X = \langle x_0, x_1, \dots, x_t \rangle$. We'll find this view useful later on, when we analyze the error rate of the Miller-Rabin primality test. Note that if $x_i = 1$ for some $0 \leq i < t$, WITNESS might not compute the rest of the sequence. If it were to do so, however, each value $x_{i+1}, x_{i+2}, \dots, x_t$ would be 1, so we can consider these positions in the sequence X as being all 1s. There are four cases:

1. $X = \langle \dots, d \rangle$, where $d \neq 1$: the sequence X does not end in 1. Return TRUE in line 8, since a is a witness to the compositeness of n (by Fermat's Theorem).
2. $X = \langle 1, 1, \dots, 1 \rangle$: the sequence X is all 1s. Return FALSE, since a is not a witness to the compositeness of n .
3. $X = \langle \dots, -1, 1, \dots, 1 \rangle$: the sequence X ends in 1, and the last non-1 is equal to -1 . Return FALSE, since a is not a witness to the compositeness of n .
4. $X = \langle \dots, d, 1, \dots, 1 \rangle$, where $d \neq \pm 1$: the sequence X ends in 1, but the last non-1 is not -1 . Return TRUE in line 6: a is a witness to the compositeness of n , since d is a nontrivial square root of 1.

Now, let's examine the Miller-Rabin primality test based on how it uses the WITNESS procedure. As before, assume that n is an odd integer greater than 2.

The procedure MILLER-RABIN is a probabilistic search for a proof that n is composite. The main loop (beginning on line 1) picks up to s random values of a from $\mathbb{Z}_n^+$, except for 1 and $n - 1$ (line 2). If it picks a value of a that is a witness to the compositeness of n , then MILLER-RABIN returns COMPOSITE on line 4. Such a result is always correct, by the correctness of WITNESS. If MILLER-RABIN finds no witness in s trials, then the procedure assumes that it found no witness because no witnesses exist, and therefore it assumes that n is prime. We'll see that this result is likely to be correct if s is large enough, but there is still a tiny chance that the procedure could be unlucky in its choice of s random values of a , so that even though the procedure failed to find a witness, at least one witness exists.

To illustrate the operation of MILLER-RABIN, let n be the Carmichael number 561, so that $n - 1 = 560 = 2^4 \cdot 35$, $t = 4$, and $u = 35$. If the procedure chooses $a = 7$ as a base, the column for $b = 35$ in Figure 31.4 (Section 31.6) shows that WITNESS computes $x_0 = a^{35} = 241 \pmod{561}$. Because of how the MODULAR-EXPONENTIATION procedure operates recursively on its parameter b , the first four columns in Figure 31.4 represent the factor 2^4 of 560—the rightmost four zeros in the binary representation of 560—reading these four zeros from right to left in the binary representation. Thus WITNESS computes the sequence $X = \langle 241, 298, 166, 67, 1 \rangle$. Then, in the last squaring step, WITNESS discovers that a^{280} is a nontrivial square root of 1 since $a^{280} = 67 \pmod{n}$ and $(a^{280})^2 = a^{560} = 1 \pmod{n}$. Therefore, $a = 7$ is a witness to the compositeness of n , WITNESS(7, n) returns TRUE, and MILLER-RABIN returns COMPOSITE.

If n is a β -bit number, MILLER-RABIN requires $O(s\beta)$ arithmetic operations and $O(s\beta^3)$ bit operations, since it requires asymptotically no more work than s modular exponentiations.

Error rate of the Miller-Rabin primality test

If MILLER-RABIN returns PRIME, then there is a very slim chance that it has made an error. Unlike PSEUDOPRIME, however, the chance of error does not depend on n : there are no bad inputs for this procedure. Rather, it depends on the size of s and the “luck of the draw” in choosing base values a . Moreover, since each test is more stringent than a simple check of equation (31.39), we can expect on general principles that the error rate should be small for randomly chosen integers n . The following theorem presents a more precise argument.

Theorem 31.39

If n is an odd composite number, then the number of witnesses to the compositeness of n is at least $(n - 1)/2$.

Proof The proof shows that the number of nonwitnesses is at most $(n - 1)/2$, which implies the theorem.

We start by claiming that any nonwitness must be a member of $\mathbb{Z}_n^*$. Why? Consider any nonwitness a . It must satisfy $a^{n-1} = 1 \pmod{n}$ or, equivalently, $a \cdot a^{n-2} = 1 \pmod{n}$. Thus the equation $ax = 1 \pmod{n}$ has a solution, namely a^{n-2} . By Corollary 31.21 on page 924, $\gcd(a, n) \mid 1$, which in turn implies that $\gcd(a, n) = 1$. Therefore, a is a member of $\mathbb{Z}_n^*$, and all nonwitnesses belong to $\mathbb{Z}_n^*$.

To complete the proof, we show that not only are all nonwitnesses contained in $\mathbb{Z}_n^*$, they are all contained in a proper subgroup B of $\mathbb{Z}_n^*$ (recall that B is a *proper* subgroup of $\mathbb{Z}_n^*$ when B is subgroup of $\mathbb{Z}_n^*$ but B is not equal to $\mathbb{Z}_n^*$). By Corollary 31.16 on page 921, we then have $|B| \leq |\mathbb{Z}_n^*|/2$. Since $|\mathbb{Z}_n^*| \leq n - 1$, we obtain $|B| \leq (n - 1)/2$. Therefore, if all nonwitnesses are contained in a proper subgroup of $\mathbb{Z}_n^*$, then the number of nonwitnesses is at most $(n - 1)/2$, so that the number of witnesses must be at least $(n - 1)/2$.

To find a proper subgroup B of $\mathbb{Z}_n^*$ containing all of the nonwitnesses, we consider two cases.

Case 1: There exists an $x \in \mathbb{Z}_n^*$ such that

$$x^{n-1} \neq 1 \pmod{n}.$$

In other words, n is not a Carmichael number. Since, as noted earlier, Carmichael numbers are extremely rare, case 1 is the more typical case (e.g., when n has been chosen randomly and is being tested for primality).

Let $B = \{b \in \mathbb{Z}_n^* : b^{n-1} = 1\}$. The set B must be nonempty, since $1 \in B$. The set B is closed under multiplication modulo n , and so B is a subgroup of $\mathbb{Z}_n^*$ by Theorem 31.14. Every nonwitness belongs to B , since a nonwitness a satisfies $a^{n-1} = 1 \pmod{n}$. Since $x \in \mathbb{Z}_n^* - B$, we have that B is a proper subgroup of $\mathbb{Z}_n^*$.

Case 2: For all $x \in \mathbb{Z}_n^*$,

$$x^{n-1} = 1 \pmod{n}. \tag{31.40}$$

In other words, n is a Carmichael number. This case is extremely rare in practice. Unlike a pseudoprimality test, however, the Miller-Rabin test can efficiently determine that Carmichael numbers are composite, as we're about to see.

In this case, n cannot be a prime power. To see why, suppose to the contrary that $n = p^e$, where p is a prime and $e > 1$. We derive a contradiction as follows. Since we assume that n is odd, p must also be odd. Theorem 31.32 on page 933 implies that $\mathbb{Z}_n^*$ is a cyclic group: it contains a generator g such that $\text{ord}_n(g) = |\mathbb{Z}_n^*| = \phi(n) = p^e(1 - 1/p) = (p - 1)p^{e-1}$. (The formula for $\phi(n)$ comes from equation (31.21) on page 920.) By equation (31.40), we have $g^{n-1} = 1 \pmod{n}$. Then the discrete logarithm theorem (Theorem 31.33 on page 933, taking $y = 0$) implies that $n - 1 = 0 \pmod{\phi(n)}$, or

$$(p - 1)p^{e-1} \mid p^e - 1.$$

This statement is a contradiction for $e > 1$, since $(p - 1)p^{e-1}$ is divisible by the prime p , but $p^e - 1$ is not. Thus n is not a prime power.

Since the odd composite number n is not a prime power, we decompose it into a product $n_1 n_2$, where n_1 and n_2 are odd numbers greater than 1 that are relatively prime to each other. (There may be several ways to decompose n , and it does not matter which one we choose. For example, if $n = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r}$, then we can choose $n_1 = p_1^{e_1}$ and $n_2 = p_2^{e_2} p_3^{e_3} \cdots p_r^{e_r}$.)

Recall that t and u are such that $n - 1 = 2^t u$, where $t \geq 1$ and u is odd, and that for an input a , the procedure WITNESS computes the sequence

$$X = \langle a^u, a^{2u}, a^{2^2 u}, \dots, a^{2^t u} \rangle$$

where all computations are performed modulo n .

Let us call a pair (v, j) of integers *acceptable* if $v \in \mathbb{Z}_n^*$, $j \in \{0, 1, \dots, t\}$, and

$$v^{2^j u} = -1 \pmod{n}.$$

Acceptable pairs certainly exist, since u is odd. Choose $v = n - 1$ and $j = 0$, and let $u = 2k + 1$, so that $v^{2^j u} = (n - 1)^u = (n - 1)^{2k+1}$. Taking this number modulo n gives $(n - 1)^{2k+1} = (n - 1)^{2k} \cdot (n - 1) = (-1)^{2k} \cdot -1 = -1 \pmod{n}$. Thus, $(n - 1, 0)$ is an acceptable pair. Now pick the largest possible j such that there exists an acceptable pair (v, j) , and fix v so that (v, j) is an acceptable pair. Let

$$B = \{x \in \mathbb{Z}_n^* : x^{2^j u} = \pm 1 \pmod{n}\}.$$

Since B is closed under multiplication modulo n , it is a subgroup of $\mathbb{Z}_n^*$. By Theorem 31.15 on page 921, therefore, $|B|$ divides $|\mathbb{Z}_n^*|$. Every nonwitness must be a member of B , since the sequence X produced by a nonwitness must either be all 1s or else contain a -1 no later than the j th position, by the maximality of j . (If (a, j') is acceptable, where a is a nonwitness, we must have $j' \leq j$ by how we chose j .)

We now use the existence of v to demonstrate that there exists a $w \in \mathbb{Z}_n^* - B$, and hence that B is a proper subgroup of $\mathbb{Z}_n^*$. Since $v^{2^j u} = -1 \pmod{n}$, we also have $v^{2^j u} = -1 \pmod{n_1}$ by Corollary 31.29 to the Chinese remainder theorem. By Corollary 31.28, there exists a w simultaneously satisfying the equations

$$w = v \pmod{n_1},$$

$$w = 1 \pmod{n_2}.$$

Therefore,

$$w^{2^j u} = -1 \pmod{n_1},$$

$$w^{2^j u} = 1 \pmod{n_2}.$$

Corollary 31.29 gives that $w^{2^j u} \neq 1$ implies $w^{2^j u} \neq -1$ and also that $w^{2^j u} \neq -1$ implies $w^{2^j u} \neq -1 \pmod{n}$. Hence, we conclude that $w^{2^j u} \neq \pm 1 \pmod{n}$, and so $w \notin B$.

It remains to show that $w \in \mathbb{Z}_n^*$. We start by working separately modulo n_1 and modulo n_2 . Working modulo n_1 , since $v \in \mathbb{Z}_n^*$, we have that $\gcd(v, n) = 1$. Also, we have $\gcd(v, n_1) = 1$, since if v does not have any common divisors with n , then it certainly does not have any

common divisors with n_1 . Since $w = v \pmod{n_1}$, we see that $\gcd(w, n_1) = 1$. Working modulo n_2 , we have $w = 1 \pmod{n_2}$ implies $\gcd(w, n_2) = 1$ by Exercise 31.2-3. Since $\gcd(w, n_1) = 1$ and $\gcd(w, n_2) = 1$, Theorem 31.6 on page 908 yields $\gcd(w, n_1 n_2) = \gcd(w, n) = 1$. That is, $w \in \mathbb{Z}_n^*$.

Therefore, we have $w \in \mathbb{Z}_n^* - B$, and we can conclude in case 2 that B , which includes all nonwitnesses, is a proper subgroup of $\mathbb{Z}_n^*$ and therefore has size at most $(n - 1)/2$.

In either case, the number of witnesses to the compositeness of n is at least $(n - 1)/2$. ■

Theorem 31.40

For any odd integer $n > 2$ and positive integer s , the probability that $\text{MILLER-RABIN}(n, s)$ errs is at most 2^{-s} .

Proof By Theorem 31.39, if n is composite, then each execution of the **for** loop of lines 1–4 of MILLER-RABIN has a probability of at least $1/2$ of discovering a witness to the compositeness of n . MILLER-RABIN makes an error only if it is so unlucky as to miss discovering a witness to the compositeness of n on each of the s iterations of the main loop. The probability of such a sequence of misses is at most 2^{-s} . ■

If n is prime, MILLER-RABIN always reports **PRIME**, and if n is composite, the chance that MILLER-RABIN reports **PRIME** is at most 2^{-s} .

When applying MILLER-RABIN to a large randomly chosen integer n , however, we need to consider as well the prior probability that n is prime, in order to correctly interpret MILLER-RABIN 's result. Suppose that we fix a bit length β and choose at random an integer n of length β bits to be tested for primality, so that $\beta \approx \lg n \approx 1.443 \ln n$. Let A denote the event that n is prime. By the prime number theorem (Theorem 31.37), the probability that n is prime is approximately

$$\Pr \{A\} \approx 1/\ln n$$

$$\approx 1.443/\beta.$$

Now let B denote the event that MILLER-RABIN returns PRIME. We have that $\Pr\{\overline{B} \mid A\} = 0$ (or equivalently, that $\Pr\{B \mid A\} = 1$) and $\Pr\{B \mid \overline{A}\} \leq 2^{-s}$ (or equivalently, that $\Pr\{\overline{B} \mid \overline{A}\} > 1 - 2^{-s}$).

But what is $\Pr\{A \mid B\}$, the probability that n is prime, given that MILLER-RABIN has returned PRIME? By the alternate form of Bayes's theorem (equation (C.20) on page 1189) and approximating $\Pr\{B \mid \overline{A}\}$ by 2^{-s} , we have

$$\begin{aligned}\Pr\{A \mid B\} &= \frac{\Pr\{A\} \Pr\{B \mid A\}}{\Pr\{A\} \Pr\{B \mid A\} + \Pr\{\overline{A}\} \Pr\{B \mid \overline{A}\}} \\ &\approx \frac{(1/\ln n) \cdot 1}{(1/\ln n) \cdot 1 + (1 - 1/\ln n) \cdot 2^{-s}} \\ &\approx \frac{1}{1 + 2^{-s}(\ln n - 1)}.\end{aligned}$$

This probability does not exceed $1/2$ until s exceeds $\lg(\ln n - 1)$. Intuitively, that many initial trials are needed just for the confidence derived from failing to find a witness to the compositeness of n to overcome the prior bias in favor of n being composite. For a number with $\beta = 1024$ bits, this initial testing requires about

$$\begin{aligned}\lg(\ln n - 1) &\approx \lg(\beta/1.443) \\ &\approx 9\end{aligned}$$

trials. In any case, choosing $s = 50$ should suffice for almost any imaginable application.

In fact, the situation is much better. If you are trying to find large primes by applying MILLER-RABIN to large randomly chosen odd integers, then choosing a small value of s (say 3) is unlikely to lead to erroneous results, though we won't prove it here. The reason is that for a randomly chosen odd composite integer n , the expected number of nonwitnesses to the compositeness of n is likely to be considerably smaller than $(n - 1)/2$.

If the integer n is not chosen randomly, however, the best that can be proven is that the number of nonwitnesses is at most $(n - 1)/4$, using an

improved version of Theorem 31.39. Furthermore, there do exist integers n for which the number of nonwitnesses is $(n - 1)/4$.

Exercises

31.8-1

Prove that if an odd integer $n > 1$ is not a prime or a prime power, then there exists a nontrivial square root of 1, modulo n .

★ 31.8-2

It is possible to strengthen Euler's theorem (Theorem 31.30) slightly to the form

$$a^{\lambda(n)} = 1 \pmod{n} \text{ for all } a \in \mathbb{Z}_n^*,$$

where $n = p_1^{e_1} \cdots p_r^{e_r}$ and $\lambda(n)$ is defined by

$$\lambda(n) = \text{lcm}(\phi(p_1^{e_1}), \dots, \phi(p_r^{e_r})).$$

Prove that $\lambda(n) \mid \phi(n)$. A composite number n is a Carmichael number if $\lambda(n) \mid n - 1$. The smallest Carmichael number is $561 = 3 \cdot 11 \cdot 17$, for which $\lambda(n) = \text{lcm}(2, 10, 16) = 80$, which divides 560. Prove that Carmichael numbers must be both “square-free” (not divisible by the square of any prime) and the product of at least three primes. (For this reason, they are not common.)

31.8-3

Prove that if x is a nontrivial square root of 1, modulo n , then $\gcd(x - 1, n)$ and $\gcd(x + 1, n)$ are both nontrivial divisors of n .

Problems

31-1 Binary gcd algorithm

Most computers can perform the operations of subtraction, testing the parity (odd or even) of a binary integer, and halving more quickly than computing remainders. This problem investigates the *binary gcd algorithm*, which avoids the remainder computations used in Euclid's algorithm.